

Predicting English Premier League Match Outcomes Using Machine Learning

Mani Tousi-Panah

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Abstract

This report describes the design, implementation, and evaluation of a machine learning model, Crossbar, that predicts match outcomes in the English Premier League. It documents the data sources, feature engineering process, model selection, training procedure, and performance analysis.

1 Introduction

This project was completed by **Mani Tousi-Panah**, a second-year undergraduate student in the Department of Mathematics and Computation Science at the University of Toronto Mississauga. I am currently majoring in Mathematics and Statistics with a minor in Computer Science. My academic background includes coursework in integral calculus, convergence and divergence of series, introductory probability and mathematical modelling, linear algebra, probability and inductive logic, Probability and Statistics I, and introductory Python programming. Through these courses I have developed foundational skills in mathematical reasoning, statistical thinking, and computational problem solving, all of which play a central role in the design and analysis of data-driven models.

My interest in football analytics and predictive modelling motivated the development of this project. Rather than approaching the work purely as a technical exercise, I viewed it as an opportunity to apply theoretical concepts from mathematics, probability, and statistics to a real-world domain. Throughout the project, I made use of a range of online learning resources, including documentation, tutorials, academic discussions, and artificial intelligence tools, which supported both the software implementation and the conceptual understanding of the modelling pipeline. These resources were used not as a replacement for learning, but as an aid in exploring ideas, validating approaches, and refining my thinking throughout the research process.

My broader academic and career interests lie in quantitative analysis, data science, and applied statistical modelling, and projects of this nature serve as an important bridge between formal coursework and independent research practice. This study therefore reflects both a technical exploration of football match prediction and a personal effort to deepen my skills in modelling, experimentation, and data-driven inference within a mathematically grounded framework.

2 Data Collection and Preprocessing

The dataset used in this project was obtained from the public football statistics archive maintained at <https://www.football-data.co.uk/englandm.php>. This resource provides historical English Premier League match results in CSV format, including full-time scores, team names, match dates, and a variety of bookmaker odds and statistical indicators. For each season, one file corresponds to all fixtures played during that campaign. The raw data files were downloaded for multiple recent seasons and stored in the `data/raw` directory of the project.

After collection, the seasonal files were concatenated into a unified dataset and assigned a `SeasonFile` identifier to preserve source traceability. Basic quality checks were then applied, including the removal of rows containing missing or corrupted fields relevant to model training. Since the prediction task focuses on full-time outcomes, only columns associated with the match result and selected feature variables were retained.

The target variable (`FTR`) encodes the match outcome using the convention provided in the source files: “H” for home win, “D” for draw, and “A” for away win. For modelling purposes, this categorical label was mapped to an integer representation, with 0 denoting a home win, 1 a draw, and 2 an away win. This encoding enables the use of multiclass classification algorithms while preserving the semantic structure of the outcome space.

Feature preprocessing consisted of selecting and transforming variables required by the modelling pipeline. Initial features were derived from Bet365 bookmaker odds (B365H, B365D, B365A), which were converted into implied probabilities and normalized to form a consistent probability distribution across outcomes. Additional football-informed features were generated through rolling averages, form-based statistics, goal differentials, and season-aggregate measures. These features were constructed programmatically during the feature-engineering stage rather than manually embedded in the raw data.

Finally, the cleaned and engineered dataset was split into training and test partitions using stratified sampling to preserve the class distribution across splits. This preprocessing workflow ensures that the model is trained on reliable, consistently structured historical data while maintaining reproducibility across experimental runs.

3 Feature Engineering

3.1 Feature Construction

For the baseline model, the feature set consists of the 1X2 match odds published by the Bet365 sportsbook. These variables represent the implied likelihood of each possible outcome: B365H (home win), B365D (draw), and B365A (away win). Since betting odds encode market expectations regarding match performance, they provide a strong initial predictive signal. The target variable (`Result`) is encoded as 0 for home wins, 1 for draws, and 2 for away wins. After removing rows with missing odds, the dataset contains 1140 matches and 3 numerical features.

B365H	B365D	B365A	Result
4.20	3.60	1.85	2
11.00	6.00	1.25	1
3.75	3.50	2.00	0
2.25	3.40	3.20	0
1.66	3.80	5.25	0

Table 1: Example rows from the modeling dataset showing Bet365 odds and encoded outcome labels. Lower odds indicate a higher bookmaker-implied probability for that outcome.

Table 1 shows an example subset of the dataset used for model training. Each row corresponds to a single match, with the Bet365 odds for a home win, draw, and away win, together with the encoded outcome label.

For example, taking a close look at the first row in the table, Bet365 thought the **away team was the favorite**, and indeed the match result was **2 (away win)**.

3.2 Train-Test Splitting

The dataset is divided into training and testing partitions using an 80/20 split, stratified by the class label to preserve the original distribution of match outcomes. The training set (912 matches) is used to fit the model, while the testing set (228 matches) is reserved for evaluation. This ensures that model performance is assessed on unseen data, providing a more realistic estimate of predictive accuracy.

3.3 Baseline Model Performance

3.3.1 Bet 365 odds feature

Class	Precision	Recall	F1-score	Support
0 (Home win)	0.57	0.86	0.69	103
1 (Draw)	0.00	0.00	0.00	52
2 (Away win)	0.54	0.53	0.54	73
Accuracy	—	—	0.56	228
Macro avg	0.37	0.47	0.41	228
Weighted avg	0.43	0.56	0.48	228

Table 2: Classification performance of the baseline logistic regression model on the test set using Bet365 odds as features.

Earlier, we computed the naive baseline, which was a home win about 45% of the time. So our model beats this baseline by about 11 percentage points. This means the model is genuinely learning the signal from the odds. This is a good first result. Class 0 - Home Wins, Recall = 0.86, the model correctly identifies 86% of home wins. Precision, however, =

0.57, meaning when it predicts home win, it is right 57% of the time. Essentially, the model is currently strongly biased towards predicting home wins (which matches real-world prior + odds behavior). Class 1 - Draws, Recall, and Precision all = 0, this is because the model never predicts draws. This is expected because Draws are the least common outcome, odds signals for draws are weak, and Logistic regression with only odds tends to collapse away from this class. This is actually a very good learning insight. We can later try to improve this by adding match statistics, adding form features, trying different models, etc. For now, this is a realistic baseline limitation. Class 2 - Away Wins, Recall = 0.53 and Precision = 0.54, meaning the model predicts away wins moderately well. To conclude, there is clearly a lot of work to be done. Again, these are the results based on the data sets which were retrieved for 22/23, 23/24, and 24/25 seasons.

Let's improve our model slightly by turning the implied probability of the betting site into odds.

HomeTeam	AwayTeam	B365H	B365D	B365A	HomeFav	p_{home}	p_{draw}	p_{away}	GoalDiff	Result
Crystal Palace	Arsenal	4.20	3.60	1.85	0	0.226	0.263	0.512	-2	2
Fulham	Liverpool	11.00	6.00	1.25	0	0.086	0.158	0.756	0	1
Bournemouth	Aston Villa	3.75	3.50	2.00	0	0.253	0.271	0.475	2	0
Leeds	Wolves	2.25	3.40	3.20	1	0.423	0.280	0.297	1	0
Newcastle	Nott'm Forest	1.66	3.80	5.25	1	0.570	0.249	0.181	2	0

Table 3: Example rows from the engineered dataset showing Bet365 odds, derived implied probabilities, favourite indicator, and encoded match outcome. Lower odds correspond to higher bookmaker-implied probability; the probabilities are normalized to sum to one.

Class	Precision	Recall	F1-score	Support
0 (Home win)	0.58	0.82	0.68	103
1 (Draw)	0.00	0.00	0.00	52
2 (Away win)	0.52	0.60	0.56	73
Accuracy	—	—	0.56	228
Macro avg	0.37	0.47	0.41	228
Weighted avg	0.43	0.56	0.49	228

Table 4: Classification performance of the logistic regression model on the test set. The model uses implied probability features derived from Bet365 odds. Accuracy is computed across all three outcome classes (home win, draw, away win).

Although the implied probability features did not significantly change the overall predictive accuracy relative to the raw odds baseline, they provide a more stable and interpretable representation of the pricing signal. By removing the bookmaker's margin and expressing outcomes in a probability space, the feature set becomes better structured for learning and improving the foundation for subsequent football-derived features introduced in later stages of the model.

3.3.2 Rolling Form Features

We're now ready for the step that should actually improve predictive performance. Adding form/performance history features lets the model understand things like recent momentum, team strength trends, whether a match is balanced (more likely to draw), and home vs away performance differences. This adds *real football knowledge* on top of the odds signal.

First step we are going to take is to add a function to our model which can compute rolling form. This function computes, for each team, the average goals scored (last 5), average goals conceded (last 5), and average points per match (last 5), and attaches values for the home team and away team. These features offer a concise representation of recent team strength, momentum, and trends in attacking/defensive performance.

Since the table is too large to paste here, a brief example illustrates what these new features do for us. The table includes the columns `Home_PTS_roll` and `Away_PTS_roll`, which represent the average points per match over the last N games. We also have `Home_GF_roll` and `Away_GF_roll`, which correspond to rolling goals scored per match, and `Home_GA_roll` and `Away_GA_roll`, which correspond to rolling goals conceded per match. These values summarize form, momentum, and recent performance strength.

For example, consider West Ham vs. Manchester City, where West Ham is the home team and Manchester City is the away team. The model returns `Home_PTS_roll` = 1.8 and `Away_PTS_roll` = 2.0, meaning Manchester City enter the match in slightly stronger form. After adding this feature to our model and training it, we get the following table back:

Class	Precision	Recall	F1-score	Support
0 (Home win)	0.73	0.84	0.78	98
1 (Draw)	0.38	0.16	0.23	49
2 (Away win)	0.65	0.77	0.70	69
Accuracy	—	—	0.66	216
Macro avg	0.58	0.59	0.57	216
Weighted avg	0.62	0.66	0.63	216

Table 5: Classification performance of the logistic regression model using both odds-based implied probabilities and rolling form features. Accuracy improves to 66%, with noticeable gains in the away-win and draw classes.

Our old model had an accuracy of about 56%; it could not predict draws. Our new model (odds + team forms) has an accuracy of 66%. Draw class now has 38% precision and 16% recall. Away win recall improved, and Home win precision improved. This is a real performance gain. The model now understands the balance of strength situations, which are where draws happen. When both teams have similar form, similar goal rates, and similar defensive stability, the draw probability increases.

We have made substantial progress up to this point, making this an appropriate stage to generate a *confusion matrix* to evaluate the model's classification performance. The confusion matrix allows us to identify where the model is making correct predictions and where misclassifications occur. To construct it, we import the *confusion_matrix* function from the *sklearn.metrics* module in Python.

	Pred 0 (Home)	Pred 1 (Draw)	Pred 2 (Away)
True 0 (Home win)	82	6	10
True 1 (Draw)	22	8	19
True 2 (Away win)	9	7	53

Table 6: Confusion matrix for the logistic regression model with odds and form features. Rows correspond to true labels, columns to predicted labels.

From the confusion matrix, we observe that **Home wins** are predicted accurately, with 82 instances correctly classified and only 16 misclassified (6 as draws and 10 as away wins). This aligns closely with the previously reported recall of approximately 0.84.

Similarly, **Away wins** exhibit strong performance, with 53 correct predictions and 16 misclassifications, corresponding to a recall of roughly 0.77.

In contrast, the model performs poorly on **Draws**, correctly predicting only 8 out of 49 cases. Most draw outcomes are instead classified as wins for either the home or away team. This behavior is expected, as draws are relatively rare and inherently ambiguous compared to decisive outcomes. Nevertheless, this represents an improvement over earlier results, where the model failed to predict any draws.

Overall, the model demonstrates strong predictive capability for Home and Away wins, with Draws being the primary weakness. Consequently, improving draw prediction would lead to a substantial overall improvement in model performance.

3.3.3 Form Gap & Balance Features

We further improve the model by introducing an additional layer of features, namely **gap (difference) features**, which capture the relative strength of the home team versus the away team. These features typically lead to improved predictive accuracy, particularly for **draw detection**, since football match outcomes are more strongly influenced by relative team strength than by absolute performance metrics.

Specifically, we augment our Python implementation with a function that computes the following variables: **PTS_gap**, representing the difference in recent points performance; **GF_gap**, capturing the relative attacking strength; **GA_gap**, measuring the defensive advantage between the away and home teams; and **Form_balance_***, which quantifies how evenly matched the two teams are (with smaller values indicating fixtures that are more likely to result in a draw).

These features allow the model to better distinguish between **clear favourite-underdog** matchups and **balanced fixtures**, the latter of which are more draw-prone. After incorporating these features and retraining the model, we obtain the following confusion matrix.

	Pred 0 (Home)	Pred 1 (Draw)	Pred 2 (Away)
True 0 (Home win)	81	6	11
True 1 (Draw)	22	9	18
True 2 (Away win)	9	6	54

Table 7: Confusion matrix for the logistic regression model with odds, form features, and gap difference features. Rows correspond to true labels, columns to predicted labels.

Solely looking at this table, one may conclude that our new feature is insignificant; however, examining the accuracy table reveals a different story.

Class	Precision	Recall	F1-score	Support
0 (Home win)	0.72	0.83	0.77	98
1 (Draw)	0.43	0.18	0.26	49
2 (Away win)	0.65	0.78	0.71	69
Accuracy	—	—	0.67	216
Macro avg	0.60	0.60	0.58	216
Weighted avg	0.63	0.67	0.64	216

Table 8: Classification performance of the logistic regression model using both odds-based implied probabilities, rolling form features, and gap difference features. Accuracy improves to 67%, with small gains in the away-win and draw classes.

The overall accuracy increases by approximately 1%; however, the more informative results arise from the class-specific performance, particularly for draws. Performance on **Home wins** remains largely unchanged, with precision, recall, and F1-score each decreasing by only about 0.01. A similar pattern is observed for **Away wins**, whose metrics remain nearly identical to the previous model.

In contrast, the **Draw** class shows measurable improvement. Precision increases from 0.38 to 0.43, recall from 0.16 to 0.18, and the F1-score from 0.23 to 0.26. Although draws remain the most challenging outcome to predict, the model is able to correctly identify a greater number of true draw results.

Overall, the introduction of relative-strength gap features yields a small but consistent improvement in both overall accuracy and draw-class performance. The model continues to perform strongest on decisive outcomes (home and away wins), while draws remain difficult due to their lower frequency and inherent ambiguity. The confusion matrix further indicates that most draw-related errors correspond to matches classified as narrow wins rather than extreme misclassifications, suggesting that the model is capturing meaningful competitive-balance structure.

3.3.4 Season-Long Team Strength Features

Next, we further enhance the model by incorporating **team strength priors**, which serve as season-long baseline indicators of performance. Specifically, we include rolling season-average points per game (PPG), long-term scoring strength, and an overall team rating

that is independent of short-term form. These priors help mitigate noise arising from small-sample fluctuations in recent form and provide a more stable representation of underlying team quality.

Incorporating season-long strength features is expected to provide the model with the following benefits:

- A stable estimate of overall team quality that is not overly influenced by short-term form (e.g., the most recent five matches).
- Improved predictive performance early in the season, when recent-form features tend to be particularly noisy.
- Richer contextual information for draw predictions:
 - Matches between two strong teams may indicate a higher likelihood of a draw.
 - Matchups between strong and weak teams are more likely to result in decisive outcomes.

Accordingly, we expect overall accuracy to improve, preferably exceeding 67%, alongside a modest increase in the draw-class F1-score and a confusion matrix exhibiting fewer extreme misclassifications. After implementing these features and retraining the model, we obtain the following accuracy metrics.

Class	Precision	Recall	F1-score	Support
0 (Home win)	0.78	0.85	0.81	98
1 (Draw)	0.55	0.24	0.34	49
2 (Away win)	0.67	0.84	0.74	69
Accuracy	—	—	0.71	216
Macro avg	0.66	0.64	0.63	216
Weighted avg	0.69	0.71	0.68	216

Table 9: Classification performance of the logistic regression model using both odds-based implied probabilities, rolling form features, gap difference features, and team strength priors. Accuracy improves to 71%.

Table 9 reports the per-class precision, recall and F_1 -scores for the final logistic regression model incorporating odds-based features, rolling-form statistics, gap features, and season-long team strength estimates. The model achieves an overall accuracy of 0.71, representing a substantial improvement over the earlier baseline models that used only odds and short-term form.

The class-wise metrics highlight different predictive regimes. Home wins (class 0) exhibit the strongest performance, with precision 0.78 and recall 0.85. This indicates that the model not only correctly identifies the majority of true home wins, but also that most predicted home wins correspond to matches that in fact resulted in a home victory. This behaviour aligns with established empirical patterns in football, where home advantage and team-strength asymmetry contribute to outcome stability.

Away wins (class 2) also demonstrate strong predictive quality, with recall 0.84 and an F_1 -score of 0.74. The improvement in away-win recall relative to earlier model versions suggests that the season-strength features capture persistent team quality more effectively than short-term form alone, reducing misclassification of strong away teams as underdogs.

Draws (class 1) remain the most difficult outcome to classify, reflecting their lower frequency and structural ambiguity. Nevertheless, draw precision improves to 0.55 and the F_1 -score rises to 0.34, representing a meaningful increase compared to the previous models. This indicates that although draws are still under-represented in recall, predicted draws are now more selective and correspond more frequently to truly balanced fixtures. In effect, the model learns that matches between teams of similar seasonal strength and comparable form are significantly more likely to result in stalemates.

	Pred 0 (Home)	Pred 1 (Draw)	Pred 2 (Away)
True 0 (Home win)	83	4	11
True 1 (Draw)	19	12	18
True 2 (Away win)	5	6	58

Table 10: Confusion matrix for the logistic regression model with odds, form features, and gap difference features. Rows correspond to true labels, columns to predicted labels.

Table 10 presents the confusion matrix for the same model. The diagonal cells correspond to correctly classified outcomes, while the off-diagonal entries indicate prediction errors.

The first row shows that 83 of the home wins are correctly classified, with only a small fraction misclassified as draws or away wins. This confirms the high recall values observed for class 0 in the performance table reflects the relative predictability of a strong home favourites.

The second row illustrates the behaviour of the model on draw fixtures. Although only 12 draws are classified correctly, the remaining errors tend to be distributed across the neighbouring win classes rather than indicating severe mispredictions. Most draw misclassifications occur as narrow home or away wins, which is consistent with the inherent uncertainty of balanced matches where small tactical or stochastic factors influence the outcome. Importantly, the number of correctly identified draws increases relative to earlier model versions, indicating that draw predictions are now better targeted toward genuinely equilibrated fixtures.

The third row shows that the model correctly identifies 58 of the away wins, with very few away wins incorrectly predicted as home victories. This asymmetry is desirable, as false inversions of match directionality (e.g., strong away win predicted as strong home win) represent the most severe error type in practical applications. The confusion matrix therefore, demonstrates that the model learns a structured decision boundary rather than behaving randomly across outcome classes.

From a football-analytic perspective, the results indicate that the model performs best in matches with a clear competitive imbalance. successfully distinguishing strong teams from weak ones. Draw prediction remains challenging, but improves meaningfully when incorporating season-level strength priors, suggesting that equilibrium outcomes are most

likely, when both teams exhibit similar underlying performance profiles over the course of the season.

3.3.5 Home-Only vs Away-Only Form Splits

Rather than relying solely on combined rolling averages, the model can be enhanced by incorporating **home-only** and **away-only** form splits. This allows the model to capture systematic asymmetries in team performance, such as teams that are particularly strong at home but weaker away, teams that score more frequently away on counter-attacks, or teams that defend more effectively at home while conceding more goals away. These effects are well-documented, predictive, and typically contribute to improved draw discrimination.

Implementing this extension is straightforward. Within the existing Python function *add_team_form_features*, we introduce an additional set of features that compute the home-only rolling form for the home team and the away-only rolling form for the away team.

After incorporating these features and retraining the model, the following accuracy table is obtained,

Class	Precision	Recall	F1-score	Support
0 (Home win)	0.78	0.89	0.83	93
1 (Draw)	0.65	0.24	0.35	46
2 (Away win)	0.74	0.92	0.82	66
Accuracy	—	—	0.76	205
Macro avg	0.72	0.69	0.67	205
Weighted avg	0.74	0.76	0.72	205

Table 11: Classification performance of the logistic regression model using both odds-based implied probabilities, rolling form features, gap difference features, team strength priors, and Home-Only vs Away-Only Form Splits. Accuracy improves to 76%.

After retraining with these features, overall predictive performance improved substantially. The test-set accuracy increased from approximately 0.71 in the previous configuration to 0.76, representing a meaningful gain in outcome discrimination. Table 11 shows the updated classification metrics.

The improvement is most evident in the recall and F1-scores for the home win and away win classes. The model now correctly identifies 89% of true home wins and 92% of true away wins, compared with markedly lower recall rates in earlier iterations. This indicates that the venue-specific features successfully capture persistent behavioural patterns such as home-ground familiarity, travel effects, tactical asymmetry, and environmental advantages.

The draws class remains the most difficult to predict, with a recall of 0.24. This is consistent with common findings in football analytics: draws are comparatively rare and typically arise in marginal or chaotic match conditions that are harder to capture using structured statistical features. Nonetheless, the precision for the draw class remains relatively high, meaning that when the model does predict a draw, it is often correct. The model therefore appears conservative with respect to draws, preferring to assign probability mass to more directional outcomes.

Additional insight is provided by the confusion matrix in Table 12. Most home and away outcomes are now clustered strongly along the diagonal, confirming that the feature layer enhances the model’s ability to differentiate between the dominant home fixtures and strong travelling sides.

	Pred Home	Pred Draw	Pred Away
True Home	83	3	7
True Draw	21	11	14
True Away	2	3	61

Table 12: Confusion matrix after incorporating venue-specific form features.

Overall, this feature layer demonstrates that incorporating structured, domain-aware football knowledge leads to tangible modelling gains. Rather than relying only on market odds and aggregate form, the model now encodes how performance manifests differently across match venues. This produces a classifier that is not only more accurate, but also more aligned with the strategic and behavioural dynamics of real-world football competition.

3.3.6 ELO-Based Team Strength Features

In the next stage of model development, we introduce an ELO-style dynamic team strength rating as an additional feature layer. The ELO framework originated in competitive chess and is now widely applied in sports analytics to track the evolving relative ability of teams over time. Each team is assigned a baseline rating at the start of the dataset. After every match, both teams’ ratings are updated according to the match outcome, the expected result, and the strength of the opponent. A win against a weak opponent produces a small rating increase, whereas a win against a strong opponent yields a larger adjustment. Drawn matches cause ratings to converge, particularly when teams are of dissimilar strength.

Formally, the expected result of a match between teams i and j is computed as

$$E_i = \frac{1}{1 + 10^{-(R_i - R_j)/400}},$$

where R_i and R_j denote the current ELO ratings of the two teams. After the match, the rating is updated via

$$R_i^{\text{new}} = R_i^{\text{old}} + K(S_i - E_i),$$

where S_i is the realised match outcome (1 for a win, 0.5 for a draw, 0 for a loss), and K is a sensitivity parameter. This mechanism produces a smooth, history-aware representation of team ability that evolves across the season.

Unlike short-horizon rolling statistics, which capture recent form, the ELO feature encodes a longer-term, structural estimate of team quality and competitive momentum. When incorporated alongside market odds and venue-specific form indicators, the model gains information about (i) persistent team strength, (ii) matchup balance, and (iii) situations in which neither team holds a decisive advantage. These characteristics are particularly valuable for distinguishing between closely contested fixtures and one-sided matches.

After augmenting the feature set with the three ELO-based variables (`ELO_Home`, `ELO_Away`, and their difference), the model was retrained and evaluated on the test set. The resulting classification metrics are reported in Table 13.

Class	Precision	Recall	F1-score	Support
Home win (0)	0.79	0.95	0.86	97
Draw (1)	0.68	0.40	0.50	48
Away win (2)	0.79	0.79	0.79	68
Accuracy		0.77		213
Macro avg	0.75	0.71	0.72	213
Weighted avg	0.76	0.77	0.76	213

Table 13: Classification metrics after incorporating ELO-based team strength features.

Overall accuracy increases to approximately 0.77, representing a further improvement over the previous feature configuration. The gain is most apparent in the home and away win classes. The model correctly identifies 95% of true home wins and 79% of true away wins, with both outcomes achieving strong F1-scores. This pattern suggests that the ELO ratings successfully capture persistent differences in structural team strength, enabling the classifier to better discriminate between dominant teams and weaker opponents even in the presence of short-term form variation.

The performance on the draw class also improves in a meaningful way. Although draws remain the most challenging outcome to predict, the recall rises to 0.40, indicating that the model is now more willing to assign draw outcomes in matches where the teams are of comparable strength. This behaviour is consistent with the theoretical interpretation of ELO: when team ratings converge, the likelihood of a balanced match outcome increases.

Further insight into the effect of the ELO features can be obtained from the confusion matrix shown in Table 14.

	Pred Home	Pred Draw	Pred Away
True Home	92	3	2
True Draw	17	19	12
True Away	8	6	54

Table 14: Confusion matrix for the ELO-augmented model.

The matrix exhibits a strong diagonal structure, particularly for the home and away win categories, confirming that the model reliably identifies directional outcomes when a clear strength imbalance exists. Misclassifications of draws are distributed primarily toward home or away wins, reflecting the difficulty of distinguishing finely balanced match states using observational features alone. Nevertheless, the number of correctly classified draws increases relative to earlier experiments, demonstrating that the ELO representation improves the model’s ability to recognise competitive parity between teams.

Overall, the introduction of the ELO feature layer leads to a classifier that is both more accurate and more behaviourally aligned with intuitive football dynamics. By combining short-term form, venue context, market expectations, and long-term strength evolution, the model develops a richer and more holistic understanding of match outcomes.

4 Model Design

4.1 Problem Formulation

The prediction task is formulated as a multiclass classification problem. For each match, the model predicts the full-time result, encoded as:

- Home win (H)
- Draw (D)
- Away win (A)

In the implementation, the target variable `FTR` is mapped to an integer label `Result` with values 0 (home win), 1 (draw), and 2 (away win).

4.2 Class Distribution

The target variable exhibits the following distribution in the dataset: approximately 43% home wins, 33% away wins, and 23% draws. This indicates that the classes are imbalanced, with home wins being the most frequent outcome. A baseline classifier that always predicts a home win would achieve around 43% accuracy, which serves as a reference point for evaluating the model performance.

After adding an additional season of data, the class distribution shifts slightly to approximately 45% home wins, 32% away wins, and 23% draws. This reinforces the presence of class imbalance and highlights the importance of evaluating the model against a strong baseline rather than accuracy alone.

4.3 Reliability Evaluation (Brier Score)

A meaningful step after the model has a fundamental structure with decent accuracy is to focus on probability quality, not just accuracy. Currently, the model predicts class labels well, but in order to use this in the real world, what matter is that probabilities are well calibrated. Meaning, when the model says, "*Home team has a 65% of winning*", does this actually happen 65% of the time?

To measure this, we are going to use the Brier score, which measures how accurate probabilities are (lower = better, perfect = 0).

Unlike accuracy, which evaluates whether a model predicts the correct class label, the Brier score measures the quality of the model's *predicted probabilities*. A classifier is not only expected to choose the most likely outcome, but also to express appropriate levels of confidence. The Brier score assesses how close the predicted probabilities are to the actual empirical outcomes.

4.3.1 Binary Brier Score

For binary outcomes, let $y_i \in \{0, 1\}$ denote the true label for sample i , and let $\hat{p}_i$ be the predicted probability that the event occurs. The Brier score is defined as the mean squared error between these values:

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2$$

A perfect model obtains a score of 0, while higher values indicate larger discrepancies between predicted probabilities and observed outcomes. Unlike log-loss, the Brier score is bounded between 0 and 1, which makes interpretation more intuitive.

4.3.2 Example (Binary Case)

Suppose a model assigns a probability of 0.7 for a home win. If the home team actually wins, then the contribution to the Brier score is

$$(0.7 - 1)^2 = 0.09$$

However, if the home team loses, the penalty becomes

$$(0.7 - 0)^2 = 0.49$$

Thus, confident but incorrect predictions are penalized more heavily than uncertain ones. The Brier score, therefore, rewards probability forecasts that are both accurate and *well calibrated*.

4.3.3 Multiclass Brier Score

In a multiclass setting, such as the outcomes of football matches (home win, draw, away win), The Brier score is generalized by summing the squared errors across all K classes:

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K (\hat{p}_{ik} - y_{ik})^2$$

where $\hat{p}_{ik}$ is the predicted probability of class k for sample i , and y_{ik} is a one-hot encoded indicator of the true outcome. The score, therefore, measures how much probability mass is assigned to incorrect outcomes.

This formulation has an intuitive interpretation: the model is penalized not only for predicting the wrong class, but also for assigning too much confidence in unlikely outcomes.

4.3.4 Interpretation

Low Brier scores indicate that the predicted probabilities are close to the empirical frequencies of observed outcomes. A model that predicts “home win = 0.65” for many matches should see approximately 65% of those matches result in home wins if it is well calibrated.

Conversely, a poorly calibrated model may still achieve high accuracy while producing overconfident probability estimates. The Brier score is therefore particularly important in applications where probability outputs are used for downstream decision-making, such as match forecasting, risk analysis, financial modeling, or sports analytics.

4.3.5 Reliability–Resolution Decomposition

The Brier score may be decomposed into three terms (Murphy, 1973): reliability, resolution, and uncertainty.

$$\text{Brier} = \text{Reliability} - \text{Resolution} + \text{Uncertainty}$$

Reliability measures how well predicted probabilities match observed frequencies, while resolution measures the model’s ability to separate distinct outcome regimes. This decomposition allows probability quality to be analysed beyond simple scalar error scores.

4.3.6 Football Example

Consider two models predicting the probability of a draw:

- Model A predicts 0.50 for every match.
- Model B predicts 0.20 for strongly imbalanced games and 0.40 for balanced games.

Even if both models achieve similar classification accuracy, Model B achieves a lower Brier score because its probabilities vary in a manner that reflects real competitive structure. The Brier score therefore captures information that is invisible to accuracy-based metrics.

5 Training and Evaluation

5.1 Probabilistic Evaluation Using the Brier Score

Beyond classification accuracy, it is important to evaluate the quality of the model’s probabilistic forecasts. For a multi-class outcome setting such as football match prediction, the Brier score provides a proper scoring rule that measures the mean squared difference between the predicted probability distribution and the observed outcome. For a given match with predicted probability vector $\hat{p}$ and realised outcome indicator vector y , the Brier score is defined as

$$\text{BS} = \frac{1}{K} \sum_{k=1}^K (\hat{p}_k - y_k)^2,$$

where $K = 3$ corresponds to the three possible results (home win, draw, away win). Lower values indicate better probabilistic calibration, with a perfectly calibrated model achieving a score of 0, and a model that assigns inaccurate or poorly discriminated probabilities yielding higher error.

Outcome Class	Brier Score
Home win	0.1114
Draw	0.1328
Away win	0.1023
Macro Average	0.1155

Table 15: Final Brier score values for the model on the test set.

The final version of the model was evaluated using this metric on the held-out test set. Separate Brier scores were computed for the home win, draw, and away win outcome classes, together with a macro-averaged overall score. The results are reported in Table 15.

The macro-averaged Brier score of approximately 0.12 indicates that the model produces well-calibrated probability estimates overall, meaning that predicted probabilities align closely with realised match frequencies. The lowest error is observed in the away win class (0.1023), suggesting that the model is particularly reliable when estimating the likelihood of away-team victories. This behaviour is consistent with the feature set, in which ELO ratings and venue-specific form provide strong information about strength imbalances that drive decisive away results.

The home win category also exhibits a low Brier score (0.1114), reflecting the stability of home advantage patterns in the data. The draw class yields the highest error (0.1328), which is expected given the inherent difficulty of modelling draw outcomes in football. Draws tend to arise in marginal or tactically constrained match states that are less directly captured by observable pre-match features, and are typically less frequent than win outcomes.

Taken together, the Brier analysis confirms that the model is not only accurate in terms of final predictions, but also produces probabilistically meaningful forecasts. The predicted probabilities encode useful information about outcome likelihoods rather than simply driving toward deterministic classification, making the model suitable for use in decision-oriented or risk-aware football analytics contexts.

6 Discussion and Future Work

Beyond the experimental evaluation conducted in this study, an important direction for this project is the transition from an offline modelling pipeline to a continuously updating prediction system. The long-term goal is to deploy the model in a setting where it can generate match predictions both before a season begins and dynamically as the season progresses, incorporating newly observed match results in real time.

The planned training workflow is structured around two complementary phases. Prior to the start of a new season, the model will be trained using data from the most recent five seasons. This restricted historical window is intentional: it provides sufficient behavioural history to capture stable team and league-level patterns, while avoiding the degradation that arises from including outdated performance profiles and structural changes in squad quality, management style, or tactical identity. To further mitigate temporal drift, the model will apply a weighted learning strategy in which the current and immediately preceding seasons

are assigned higher importance than older seasons, ensuring that predictions are anchored to contemporary football dynamics.

Once the season begins, the model will shift into a live updating mode. After each matchweek, the new fixtures will be added to the data table and the feature-generation pipeline will be re-computed. This enables rolling performance features, form metrics, and ELO ratings to evolve as teams gain or lose momentum. Over time, the balance of information will progressively favour current-season data, strengthening the model’s ability to reflect emerging trends such as tactical adjustments, roster changes, or performance shocks. The resulting workflow produces a hybrid forecasting system: pre-season predictions are informed by historical structure, while in-season predictions are increasingly driven by realised evidence.

Several promising extensions remain for future development. A natural next stage is the incorporation of richer match-context features, including disciplinary events (e.g., red cards), injury and availability information, match congestion effects, and potentially player-level metrics. These factors are known to influence match outcomes but require more complex and heterogeneous data sources, as well as careful modelling to avoid noise and spurious correlations. Such additions are therefore reserved for a planned “Version 2” model iteration, in which the focus will shift from team-level structural signals to finer-grained contextual and situational predictors.

Overall, the next phase of this project will transform the current model from a static analytical framework into a practical, continuously adaptive prediction engine, capable of supporting ongoing analysis and future research in football analytics.

7 Conclusion

The goal of this project was to design and evaluate a machine learning model capable of predicting the outcome of English Premier League football matches, with a particular emphasis on understanding how domain-specific features influence predictive performance and probabilistic calibration. Rather than treating the task as a purely algorithmic exercise, the modelling process was developed iteratively, with each feature layer motivated by football analytics reasoning and empirically assessed through controlled experiments.

The baseline model began with market-derived variables in the form of Bet365 pre-match odds. These features provided an initial predictive signal reflecting aggregate betting-market expectations. Subsequent feature layers introduced progressively richer representations of team performance, including rolling form statistics, home-away venue splits, and season-level strength aggregates. The addition of these features produced clear improvements in accuracy and class-wise recall, indicating that contextual performance dynamics play an important role in shaping match outcomes.

A further enhancement was achieved through the incorporation of a football-adapted ELO rating system. The ELO features provided a time-evolving estimate of structural team strength and competitive momentum, complementing short-horizon form measures and venue effects. This modification led to one of the most substantial performance gains, particularly in the correct identification of home and away wins and in the improved handling of balanced fixtures. Exploratory head-to-head matchup features were also tested, but were found to

contribute minimal additional benefit, highlighting the importance of empirical validation rather than assuming feature value a priori.

The final version of the model achieved strong predictive performance, with overall test-set accuracy of approximately 0.77 and robust precision–recall behaviour across outcome classes. To assess the quality of the probability estimates themselves, the model was additionally evaluated using the Brier score. The resulting macro-averaged Brier score of 0.1155 demonstrates that the predicted probabilities are well-calibrated and meaningfully aligned with realised match outcomes. Although draw outcomes remained the most challenging class to estimate, this behaviour is consistent with the stochastic and tactically situational nature of draws in football.

Overall, the results of this project illustrate that effective football match prediction emerges not from complex algorithms alone, but from the careful integration of domain knowledge, structured feature design, and iterative empirical evaluation. The final model provides both accurate and probabilistically credible match forecasts, and establishes a strong foundation for future extensions such as live-season updating, match-specific tactical features, or integration with decision-support applications in football analytics contexts.

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